3 NOVEMBER 2026

CLASS 02 NOTES

Rules vs learning, types of ML, and your roadmap

01	The honest answer to the AI question
02	What machine learning actually is
03	Supervised, unsupervised, reinforcement
04	The vocabulary, used correctly
05	What the two months cover, and where you go after

Machine Learning: Build → Understand → Deploy

31 live classes // 2 November – 29 December 2026

learn.rajeevshivah.me

CLASS 02 · 01

WHAT GOT CHEAP, AND WHAT GOT SCARCE

Now almost free

- Writing the model code
- Remembering library syntax
- Standard plots and boilerplate
- Explaining what an algorithm is

Scarcer than ever

- Deciding what to predict, and from what
- Spotting that a 97% score is a lie
- Choosing a split and a metric that match reality
- Knowing why the model fails, and on whom

AI makes a weak engineer worthless and a good engineer far more valuable. There is no third option, and the gap is widening every month.

CLASS 02 · 02

AI will write your model in one second. It cannot tell you *the model is wrong*.

That is the skill this course is built around. For the first two weeks I will let you use AI freely. After that, I will spend six classes showing you everything it got wrong without telling you.

CLASS 02 · 03

THE ROAD, HONESTLY MAPPED

Python→ NumPy & Pandas→ Maths intuition→ Algorithms→ Engineering judgment→ Deployment
You already have step one. We start step two in forty minutes.

Most courses stop after step four, which is exactly the part a model can now do for you. The last two steps are where the job is.

CLASS 02 · 04

WHAT MACHINE LEARNING ACTUALLY IS

Normal programming
You write the rules.

```
if "lottery" in subject:
    spam = True
elif sender in blocklist:
    spam = True
# ...and 400 more rules
# that break every week
```

Machine learning
You give examples. It finds the rules.

```
model.fit(emails, is_spam)
# the rules are now inside the
# model, learned from 50,000
# examples you never read
```

That flip is the whole idea. Everything else is detail.

CLASS 02 · 05

THE DEFINITION WE WILL USE ALL TERM

> **Machine learning is learning a function from data.** We are looking for an `f` such that `f(features) → target`, built from examples where we already know the answer.

So for every project this term, the first two questions will be the same:

1. What is the **target**? The thing I want to predict.
2. What are the **features**? The things I already know.

If you cannot answer those two, you do not have a machine learning problem yet. You have a wish.

CLASS 02 · 06

SIX WORDS, USED CORRECTLY FROM TODAY

WORD	MEANS	IN A DELIVERY DATASET
Row / sample	One observation	One delivered order
Feature	An input column	Distance, traffic, time of day
Target	The column we predict	Delivery time in minutes
Model	The learned function	The thing that maps the first to the second
Training	Finding the function	Showing it 30,000 past orders
Prediction	Using it on new data	An ETA for an order placed now

Use these words precisely. In an interview, saying "data" when you mean "features" tells them a lot about you.

CLASS 02 · 07

THREE FAMILIES

Supervised
You have the answers for past data. Learn to predict the answer for new data.

This is 90% of industry work, and almost all of this course.

Unsupervised
No answers at all. Find the structure that is already in the data.

Grouping customers, reducing 200 columns to 10.

Reinforcement
No answers, only rewards. Learn by trying, failing, and being scored.

Game-playing agents, robotics, some recommendation systems.

CLASS 02 · 08

SUPERVISED SPLITS IN TWO

Regression → a number
Delivery time in minutes. A price. A salary. Tomorrow's demand.

Algorithms: Linear Regression, Ridge, Lasso, Decision Tree, Random Forest, XGBoost

Classification → a category
Spam or not. Fraud or genuine. Will this customer leave, yes or no.

Algorithms: Logistic Regression, KNN, Naive Bayes, SVM, Decision Tree, Random Forest, XGBoost

> Notice how many names appear in both columns. Trees and boosting handle either. That is why they dominate real work and why we spend serious time on them in December.

CLASS 02 · 09

UNSUPERVISED, IN TWO JOBS

Clustering — group things that belong together

K-Means, Hierarchical clustering, DBSCAN. Used for customer segmentation, grouping similar documents, finding outliers.

Dimensionality reduction — fewer columns, same information

PCA, t-SNE. Used when you have 300 features and need 20, or when you want to draw high-dimensional data on a flat screen.

We cover K-Means and PCA in class 28, once you can handle labelled problems properly.

CLASS 02 · 10

THE TWO MONTHS, BLOCK BY BLOCK

CLASSES	BLOCK	WHAT YOU GET
1–3	Foundation	NumPy, Pandas, the workflow, your first baseline
4–8	Regression	First model, the maths under it, first live app
9–14	Classification	Logistic, trees, forests, metrics that lie, first API
15–20	*Engineering judgment*	Leakage, validation design, pipelines, error analysis
21–26	*Your own data*	Collect a dataset nobody has, XGBoost, SHAP, deploy
27–28	Text and clustering	TF-IDF, Naive Bayes, K-Means, PCA
29–31	Production and capstone	Drift, monitoring, your own project, demo day

Blocks four and five are the ones no tutorial teaches. They are also the ones interviewers ask about.

CLASS 02 · 11

WHAT YOU OWN ON 30 DECEMBER

1. **A delivery time predictor** — live on the internet, anyone can open it
2. **A loan approval classifier** — served as a real API
3. **A pull request merge predictor** — on data you collected yourself from GitHub
4. **Your own capstone** — your question, your data, your deployed app

> Four links you can paste into an application, and the ability to answer "why did you choose that metric?" without blinking. That is the deliverable.

CLASS 02 · 12

"I ALREADY KNOW NUMPY AND PANDAS"

Some of you do. I am still giving them two to three full classes, and that is deliberate.

Everything we build for the next two months sits on these two libraries. A shaky foundation here does not show up in week one. It shows up in class 20 as a bug you cannot find, in code you cannot explain.

If you are already comfortable

- Use those classes to go deeper, not to switch off
- Do the harder exercises I post instead of the basic ones
- Answer other people's doubts in the group — teaching is the fastest way to find your own gaps

> And do not worry about the advanced parts being skipped. If a topic needs more room than a class allows, **I will run extra sessions on Sundays** to cover it. Nobody gets held back, and nobody gets left behind.

CLASS 02 · 13

HOW THESE TWO MONTHS WILL RUN

DAYS	Monday, Tuesday, Thursday, Friday
TIME	8:30 PM, 90 minutes (tonight is 60)
OFF DAYS	Wednesday, Saturday, Sunday — that is your practice time, not a holiday
TOTAL	31 classes, 2 November to 30 December
EXTRA	Sunday sessions when a topic needs more room

> **The meeting link lives on the platform, never in WhatsApp.** Log in before every class and you will find it there. I will not post it in the group, so please do not ask.

Be on time. We start sharp, and the first ten minutes are usually the part that makes the rest make sense.

CLASS 02 · 14

RECORDINGS, NOTES AND CODE

1. **Every class is recorded.** If you miss one, you have not lost it.
2. **The recording, notes and code go on the platform** after each class — same place you get the meeting link.
3. **WhatsApp is for announcements and doubts only.** No links, no recordings there.

> One honest warning about recordings. Everyone who falls behind says the same sentence: "I will catch up on the recordings." Almost nobody does. The recordings are a safety net for the class you genuinely cannot attend, not a plan.

CLASS 02 · 15

BEFORE TOMORROW'S CLASS

Tomorrow is hands-on from minute one. Come with this working.

Option A — on your laptop

```
pip install numpy pandas matplotlib scikit-learn jupyter
```

Then run `jupyter notebook` and check it opens.

Option B — nothing to install

Use **Google Colab** at colab.research.google.com. Free, runs in the browser, works on any machine. Perfectly fine for the next two weeks.

Test it works

```
import numpy as np
import pandas as pd
print(np.__version__, pd.__version__)
```

> If that errors, post a screenshot in the group tonight. Do not wait until 8:30 tomorrow — we will not stop the class to fix installations.

CLASS 02 · 16

AFTER THESE TWO MONTHS — PICK YOUR ROAD

Everything below starts from what we build here. Classic ML is the common base under all of it.

ROAD	WHAT YOU ADD AFTER THIS
Data Analyst fastest first job	SQL properly, Excel, a dashboard tool (Power BI or Tableau), basic statistics, business communication
Data Scientist	Statistics and probability, experiment design and A/B testing, deeper feature work, framing business problems
ML Engineer	Software engineering, APIs, Docker, a cloud provider, serving models at scale, system design
MLOps	CI/CD, pipelines, model registry, monitoring and drift, infrastructure as code
Deep Learning / GenAI	PyTorch, neural networks, transformers, building on top of LLMs

> Notice what is missing from every row: none of them say "learn more algorithms". They say **go deeper in one direction**. The classic ML we do here is what you stand on to do that.

CLASS 02 · 17

TONIGHT, AND WHAT HAPPENS NEXT

1. **Tomorrow, 8:30 PM — NumPy.** Hands-on from the first minute. Laptop open, notebook ready.
2. **Do the setup tonight.** Not tomorrow evening.
3. **Say hello in the group** — your name, your year or your work, and what you want out of these two months.

> One ask, and I will only say it once. If you attend live and never practise, this will not work. Four classes a week and three practice days is the design. The practice days are not spare time — they are where the learning actually happens.

See you tomorrow at 8:30.
